

Name: _____

Date: _____

Problem 1.

From the truth table below, write the Boolean expression for output Y as a sum of products (one product term for each row where $Y = 1$), then draw its logic-gate circuit.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Problem 2.

From the truth table below, write the Boolean expression for output Y as a sum of products (one product term for each row where $Y = 1$), then draw its logic-gate circuit.

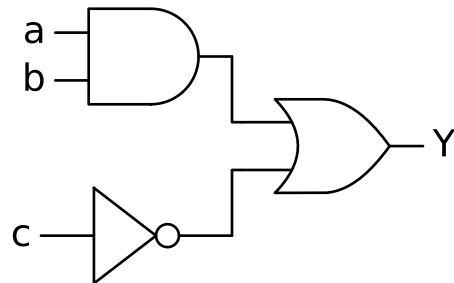
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Problem 3.

From the truth table below, write the Boolean expression for output Y as a sum of products (one product term for each row where $Y = 1$), then draw its logic-gate circuit.

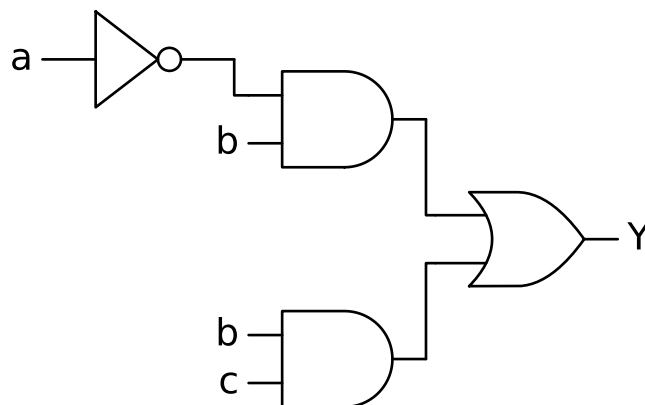
A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Problem 4.



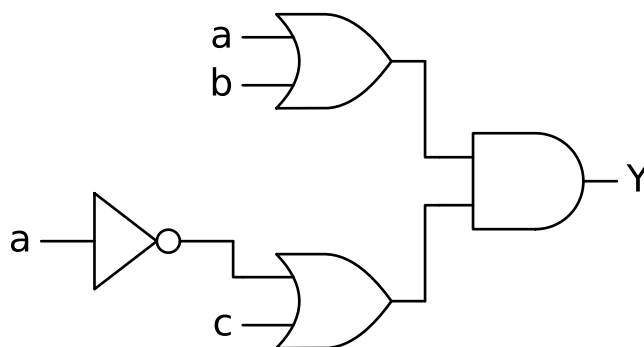
Write the Boolean expression for output Y , then draw its truth table.

Problem 5.



Write the Boolean expression for output Y , then draw its truth table.

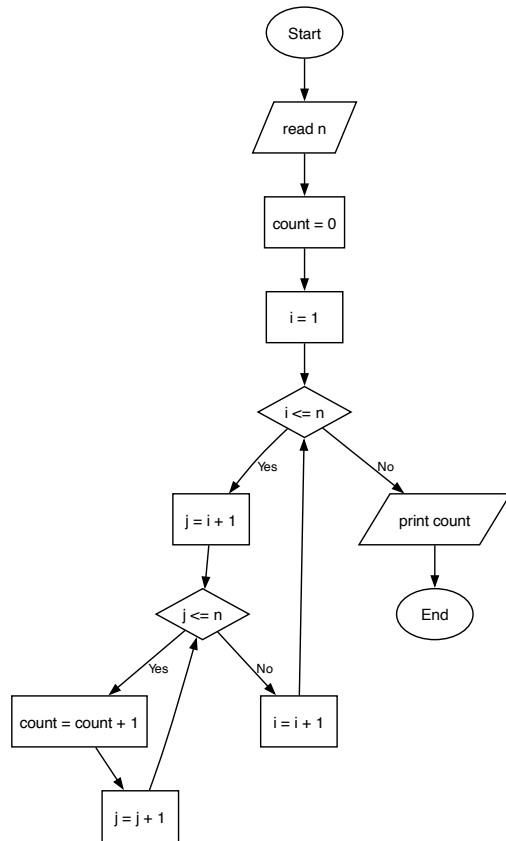
Problem 6.



Write the Boolean expression for output Y , then draw its truth table.

Problem 7.

Trace this algorithm for $n = 3$, and complete the trace table.

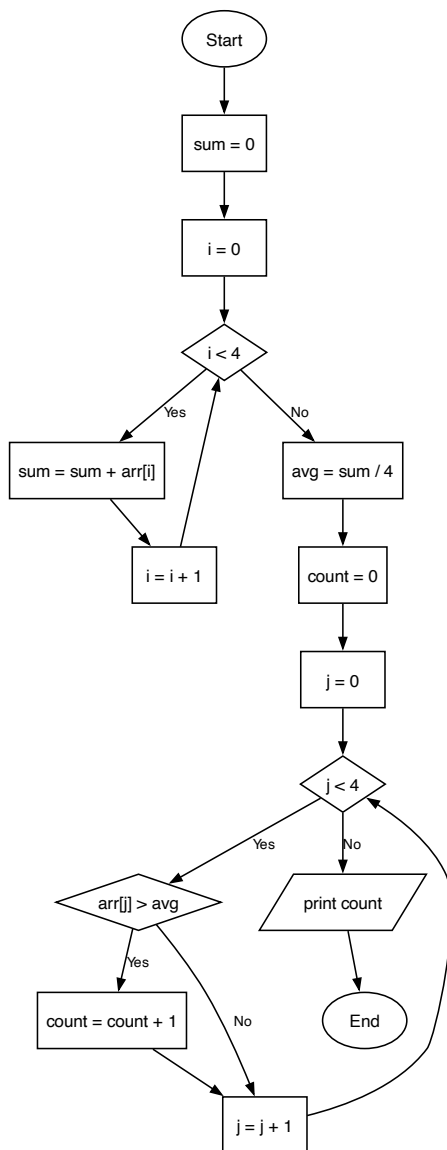
[illegible]

In the real world this is how you count the number of distinct pairs in a group of n things: the number of handshakes if n people all shake hands once, or the number of direct links in a network where every one of n computers connects to every other. It always works out to $n \times (n - 1) / 2$.

Problem 8.

Trace this algorithm, taking the list *arr* to be [70, 90, 60, 80], and complete the trace table.

Reminder: the items of a list have positions counted from 0, so the list $[60, 75, 50]$ has 3 items in positions 0, 1, 2 — $\text{arr}[i]$ means the item at position i , so $\text{arr}[0]$ is the first item.

[illegible]

This two-pass pattern — go through the data once to measure something overall, then go through it again to compare each item against that measure — is everywhere in data analysis: for example, finding how many students scored above the class average, or how many days this month were hotter than usual.